

✂ Active Directory Secure Channel Failure After Account Reset

Environment: dubz-vault.corp

Role: POS Workstation (Physical Machine)

Severity: Authentication Failure

Layer: Identity / Domain Trust

Problem Statement

When attempting to log into the POS workstation using domain credentials (RDP or console), the following error appeared:

“The security database on the server does not have a computer account for this workstation trust relationship.”

The workstation was previously domain joined and functioning normally.

Troubleshooting Flow

This guide follows a layered troubleshooting model:

1. Network Layer
 2. DNS Layer
 3. Active Directory Object Validation
 4. Secure Channel Validation
 5. Repair Procedure
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Network Layer Validation

Before assuming AD corruption, confirm DC reachability.

Test DC Connectivity

```
Test-NetConnection 10.10.10.200 -Port 53
```

```
Test-NetConnection 10.10.10.200 -Port 88
Test-NetConnection 10.10.10.200 -Port 389
Test-NetConnection 10.10.10.200 -Port 445
```

Expected:

```
TcpTestSucceeded : True
```

If **False** → investigate firewall (pfSense inter-VLAN rules).

2 DNS Validation

Confirm DNS Configuration

```
ipconfig /all
```

Verify:

```
DNS Servers . . . . . : 10.10.10.200
Connection-specific DNS Suffix : dubz-vault.corp
```

Validate DC Resolution

```
nslookup dubz-vault.corp
nltest /dsgetdc:dubz-vault.corp
```

If DC cannot be located → DNS misconfiguration.

3 Validate Computer Object in Active Directory

On Domain Controller:

```
Get-ADComputer POS -Properties PasswordLastSet
```

Confirm:

- Object exists
- Enabled = True
- PasswordLastSet is recent

If object does not exist → recreate and rejoin domain.

Validate Secure Channel (On Workstation)

Login locally using:

```
.\Administrator
```

Then run:

```
Test-ComputerSecureChannel -Verbose
```

If result:

```
False
```

Secure channel is broken.

Root Cause Identified

In this case:

- Network connectivity was confirmed.
- DNS was correct.
- AD object existed and was enabled.
- PasswordLastSet was recently changed.

This indicates:

The computer account was reset in Active Directory, but the workstation still had the old machine password.

When a computer account is reset in AD:

- AD immediately rotates the machine password.
- The workstation continues using the old password.
- Trust relationship fails.



Remediation Procedure

From the workstation (local admin session):

```
Test-ComputerSecureChannel -Repair -Credential dubz-vault\Administrator -  
Verbose
```

Enter domain admin credentials.

Expected output:

```
True
```

Reboot the workstation.



Post-Repair Validation

```
Test-ComputerSecureChannel
```

Should return:

```
True
```

Then test domain login via RDP:

```
dubz-vault\username
```

Authentication should succeed.



Why This Happens

Common causes:

- Resetting computer account in AD
- Restoring DC snapshot
- Rebuilding DC
- Machine offline beyond password rotation cycle (30 days default)

- Manual object deletion/recreation
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Enterprise Takeaways

- Always validate network and DNS before modifying AD.
 - Do NOT immediately unjoin/rejoin domain.
 - Repair secure channel first.
 - Resetting computer accounts changes machine passwords instantly.
 - RDP failures often surface secure channel issues before console login.
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Optional Hardening

To proactively monitor secure channel health:

```
Test-ComputerSecureChannel
```

Can be scheduled via Task Scheduler for periodic validation.